

Markscheme

Mathematics: analysis and approaches

Practice question bank

68. (a)

Consider the arithmetic sequence $7, 11, 15, \dots, 415$.

Find the number of terms in the sequence, and hence find the sum of all its terms.

$$u_n = 7 + 4(n - 1) = 415 \quad (M1)$$

$$n = 103 \quad \mathbf{A1}$$

$$S_{103} = \frac{103}{2}(7 + 415) \quad (M1)$$

$$= 21733 \quad \mathbf{A1}$$

[5 marks]